

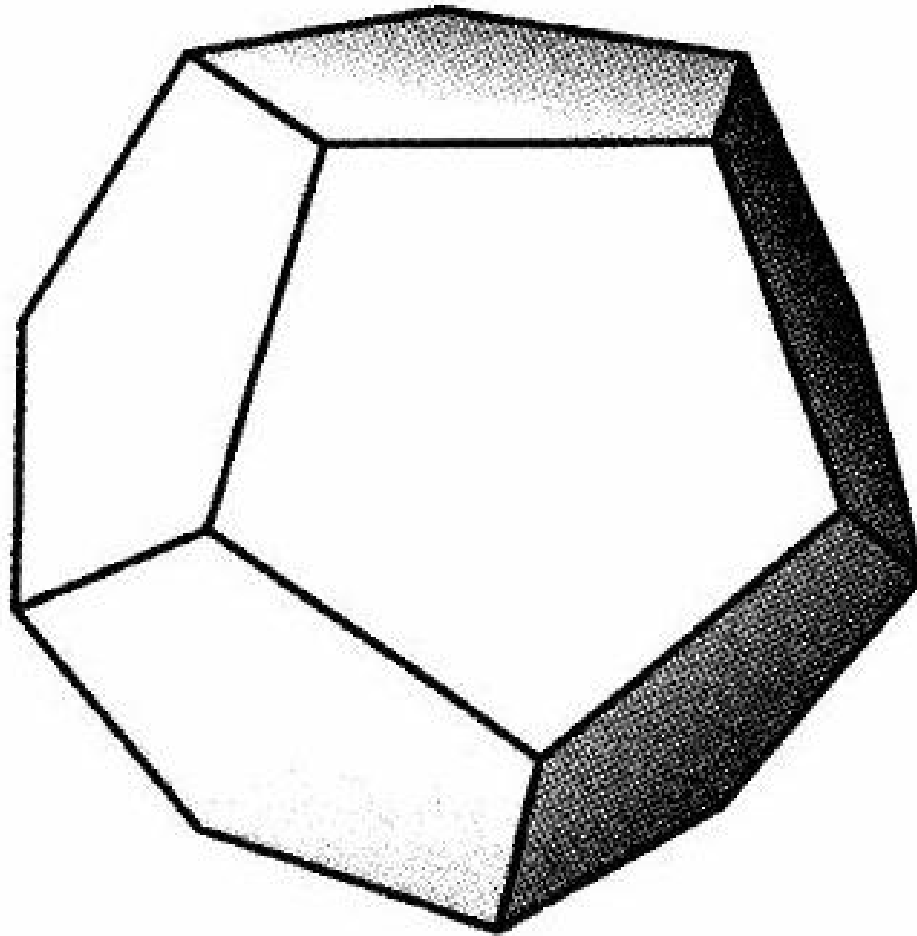
2020 AMC 10A Problem 19 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 19. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 19

Problem:

As shown in the figure below, a regular dodecahedron (the polyhedron consisting of 12 congruent regular pentagonal faces) floats in space with two horizontal faces. Note that there is a ring of five slanted faces adjacent to the top face, and a ring of five slanted faces adjacent to the bottom face. How many ways are there to move from the top face to the bottom face via a sequence of adjacent faces so that each face is visited at most once and moves are not permitted from the bottom ring to the top ring?

**Answer Choices:**

- A. 125
- B. 250
- C. 405
- D. 640
- E. 810

Solution:

Since we start at the top face and end at the bottom face without moving from the lower ring to the upper ring or revisiting a face, our journey must consist of the top face, a series of faces in the upper ring, a series of faces in the lower ring, and the bottom face, in that order.

We have 5 choices for which face we visit first on the top ring. From there, we have 9 choices for how far around the top ring we go before moving down: 1, 2, 3, or 4 faces around clockwise, 1, 2, 3, or 4 faces around counterclockwise, or immediately going down to the lower ring without visiting any other faces in the upper ring. We then have 2 choices for which lower ring face to visit first (since every upper-ring face is adjacent to exactly 2 lower-ring faces) and then once again 9 choices for how to travel around the lower ring. We then proceed to the bottom face, completing the trip.

Multiplying together all the numbers of choices we have, we get

$$5 \cdot 9 \cdot 2 \cdot 9 = \boxed{(E) 810}$$

OR

There are five ways to get to the top ring.

- **Case 1:** directly go to bottom ring For each of the 5 initial top ring faces, we have two ways of directly going to the bottom ring, as each face on the top is adjacent to two faces on the bottom.

There are $5 \cdot 2 = 10$ paths here.

- **Case 2:** traversing sideways After entering the top ring, we can also choose to travel sideways along the top ring before going to the bottom ring. For each of the 5 initial top ring faces, we can choose 2 directions to go (clockwise or counterclockwise). Once we choose a direction, we can go 1, 2, 3, or 4 faces in that direction. Notice we cannot travel 5 faces because then we would land on the original top ring face, but the problem specifies that we can only travel each face once. Once again, top ring faces are each adjacent to two bottom ring faces, so after traversing sideways we can choose 2 paths to exit.

There are $5 \cdot 2 \cdot 4 \cdot 2 = 80$ paths here.

Adding the two cases together, there are **90** ways to get to the bottom ring from the top. Next, from the bottom ring to the bottom we do casework again:

- **Case 1:** directly go to the bottom For each of the 90 ways we can get to faces on the bottom ring, there is only 1 way to get to the bottom. So there are 90 ways here.

- **Case 2:** traversing sideways For each of the 90 ways we can get to faces on the bottom ring, there are once again 2 ways to choose the direction and 4 ways to choose the amount of faces traveled. However, after traveling the faces there is only 1 way to go to the bottom face.

There are $90 \cdot 2 \cdot 4 \cdot 1 = 720$ paths here.

Adding together the cases, there are $90 + 720 = \boxed{(E) 810}$ paths.

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